

WORK EXPERIENCE

American Express

July 2023 – July 2024

Software Engineer - Associate DPM (Global Risk & Compliance Dept)

- **Saved ~\$70K** in Q1–Q2'24 through **33% data rationalization**, by designing & developing **Python** and **SQL** pipelines to resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team.
- Led the implementation of onboarding pipelines for Kabbage acquisition, migrating **~20K accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance.
- Maintained and debugged production pipelines ingesting **millions of transactions** per 5-minute Amex feed, detecting file loss and duplication through transaction-volume anomaly monitoring in **Yellowbrick**; partnered with the AML Transaction Monitoring team via **JIRA** in an Agile environment to triage and resolve data-integrity issues.

American Express

May 2022 – July 2022

Data Engineering - Analyst Intern (Credit & Fraud Risk Dept)

- Reduced development time by 80% for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios.
- Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.
- Streamlined onboarding for new data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

TECHNICAL SKILLS

Language: C, C++, Python, MATLAB, PHP, R, SQL, PostgreSQL, HiveQL, TypeScript, JavaScript**Cloud Tech:** Amazon Web Services: EC2, S3, Lambda, ECS, CodePipeline; Google Cloud : GCP, Vertex AI, Firestore.**DevOps, Infra & Delivery:** Git, Docker, Kubernetes, Apache Kafka, Redis, Linux OS, Slurm, SDLC, CI/CD, Jenkins.**App Dev & Analytics:** HTML, CSS, REST APIs, React, Node.js, FastAPI, Uvicorn, Streamlit, n8n, PHPUnit.**ML & AI:** scikit-learn, numpy, scipy, pandas, LangChain, keras, tensorflow, LangGraph, transformers.**AI Coding Tools:** Claude Code, Codex, Cursor, Github Copilot

EDUCATION

University of Michigan, Ann Arbor

Aug. 2024 – May 2026

*Master of Science in Engineering in Computer Science***GPA: 4.0/4.0**

Delhi Technological University, Delhi

July 2019 – May 2023

*Bachelor of Technology in Mathematics & Computer Science***GPA: 3.81/4.0****Coursework:** Data Structures, Algorithms, Object Oriented Programming, Software Engineering, System Design, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Natural Language Processing, Data Science, Statistics, Linear Algebra.

PUBLICATION

“University Recommendation System” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)** on 6–7th May 2023 at Gurgaon, India.

PROJECT EXPERIENCE

Wikispeech (MediaWiki extension): Open Source Contribution

- Built Wikimedia's article-to-audio download feature, enabling users to listen to and **save Wikipedia articles as speech**; delivered 4 **PHP** patches spanning schema design, **REST APIs**, JobQueue processing, and UI workflows with 40+ **PHPUnit** tests.

CineRank: Real-Time Movie Recommendation System

- Built a real-time recommendation engine that continuously updates user preferences & serves movie suggestions in **<100ms latency** on the **MovieLens 32M** dataset. Designed a streaming ML pipeline using **Kafka**, **Redis**, **PostgreSQL**, **FastAPI**, combining SVD collab filtering, cold-start fallback logic, & cached recommendation retrieval.

Needle: RAG-based Research Assistant

- Built an RA for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & uploaded PDFs, removing hallucinated citations through retrieval-backed generation. Developed a RAG pipeline using **Streamlit**, **Vertex AI**, **Gemini 2**, & **LangChain** with semantic search, citation verification, & conversational document querying.

AI Generated Text Detector: Built a logistic regression classifier using **scikit-learn** & **scipy** sparse features, with ~80% accuracy on AI-written text detection, benchmarking SGD, RMSprop, & AdamW optimizers in **pytorch**.

COVID-19 Detection using CNN: Built a **python** lung X-ray classification system using **tensorflow** and **keras** with VGG19 and ResNet50 architectures, achieving 95% accuracy on COVID-19 detection.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Generation Women Leaders 2023 & 2025 | **URAM (Micron) Scholarship** 2022 | **AWS ML Scholarship** 2021 | UCMAS International Distinction Grade 10
- **Leadership:** Judge, **MHacks 17 (MLH Official)** | **Vice Chairperson**, IEEE DTU CS | Student Advisory Board, Michigan Engineering | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & American Express, Joy of Coding (UMich)